Agent Memory Graph

Product & Deployment Proposal for LATAM Airlines

Fecha de Generación: 2026-06-29

Autor: ADO — Analytics Data Office

Asunto: Enterprise AI Agent Memory Platform — Internal Product Brief

1. Executive Summary

Agent Memory Graph (AMG) is an enterprise-grade, GCP-native AI agent coordination platform that solves the most critical failure modes of production AI deployments: context loss, hallucination, and uncontrolled inference cost. Its two core innovations — the MindMap reasoning pipeline and the Cognitive Weighting system — work together to deliver measurably better AI: lower hallucination rates, faster retrieval, and significantly reduced LLM token consumption compared to conventional RAG architectures.

The MindMap pipeline (inspired by the ACL'24 paper by Wen, Wang & Sun) forces every agent response to be built exclusively from graph-retrieved evidence. The Cognitive Weighting system (inspired by Engram's provable memory infrastructure) assigns Bayesian confidence scores to every fact in memory, applies temporal decay to stale knowledge, and gates synthesis on freshness and confidence thresholds. The practical result: agents retrieve only high-signal, recent facts; unused or contradicted facts fade out automatically; and token context is kept bounded and relevant — avoiding the sprawling, expensive context windows typical of naive RAG.

For LATAM Airlines, AMG is the missing coordination layer for the Cosmos AI platform: a persistent, shared memory fabric that all agents — from internal engineering copilots to passenger-facing chatbots — can trust and reason over. AMG is production-ready (v0.2, 843 tests passing, P99 < 5ms), deploys within LATAM's existing GCP organization, and integrates naturally with the Cosmos GenAI Gateway, DataHub, and Backstage catalog.

> Key numbers: P99 latency < 5ms (10x below SLO). Cognitive weighting bounds retrieval context to only high-confidence facts — directly reducing LLM token spend per agent turn. Grounding rate: 100% of responses backed by evidence or explicit insufficiency (zero hallucination).

2. Product Description

Agent Memory Graph is a deterministic, grounding-first memory system for coordinating multiple AI agents within a single enterprise. It eliminates hallucination by forcing every agent response to be derived from evidence retrieved from a graph database. It preserves organizational context across sessions, users, and agents. And it provides immutable audit trails for every reasoning step — critical for regulated industries and compliance-sensitive workflows.

2.1 The Problems It Solves

Enterprise AI deployments fail for consistent, well-understood reasons. AMG addresses each one with a specific architectural decision — not a workaround:

Problem	How AMG Fixes It	Key Feature
Context Loss — agents forget interactions across sessions; users re-explain the same problems repeatedly	Context tier stores full session history; Semantic tier persists organizational facts indefinitely; recall is automatic on every agent turn	Tripartite Memory Model

© 2026 LATAM Airlines

Problem	How AMG Fixes It	Key Feature
Hallucination — agents invent facts not grounded in organizational data; outputs cannot be trusted for operational or regulated decisions	MindMap pipeline: synthesis is prohibited unless facts are retrieved and serialized first. No evidence = explicit insufficiency signal, never a guess	MindMap Grounding Pipeline
High Token Cost — naive RAG floods the LLM context with loosely relevant chunks, driving up inference spend at scale	Cognitive Weighting filters the retrieval graph to only high-confidence, fresh, high-signal facts before synthesis. Bounded traversal horizons prevent context sprawl	Cognitive Weighting (Bayesian)
Low Retrieval Accuracy — vector similarity search returns topically similar but factually unreliable chunks	Graph traversal over the Semantic tier follows typed relationships between entities, producing structurally grounded paths — not approximate vector neighbors	Spanner Graph Retrieval
Auditability Gap — no reasoning traces; impossible to debug decisions or prove compliance	Audit tier logs every memory access, tool call, authorization decision, and reasoning step immutably (append-only, never deleted)	Immutable Audit Trail
Multi-Agent Coordination Breakdown — multiple agents make conflicting decisions with no shared context	All agents share the same Semantic graph within the organization. Contradiction detection flags conflicting facts before they reach synthesis	Shared Org Memory + Contradiction Detection

2.2 What AMG Provides

- Tripartite Memory Model — Strict separation of Context (user-specific session history), Semantic (durable organizational facts with Bayesian confidence scoring), and Audit (immutable compliance logs). Each tier has independent retention, mutability, and access policies.
- MindMap Reasoning Pipeline — A deterministic four-stage pipeline: Mine → Aggregate → Serialize → Synthesize. Every response is built exclusively from retrieved graph evidence. If evidence is insufficient, the system says so explicitly — no hallucination.
- Cognitive Weighting System — Bayesian log-odds weights on every node and edge, temporal decay for stale facts, and confidence + freshness gating on synthesis. Only relevant, recent, high-signal facts reach the LLM context window, reducing token cost and improving answer accuracy.
- Multi-Agent Coordination — All agents share the same Semantic memory graph. Coordinated, non-redundant operations. Contradiction detection surfaces conflicting facts before they influence a response.
- Immutable Audit Trail — Every access, tool call, and reasoning step in the Audit tier (append-only) for full compliance traceability and incident forensics.
- ABAC Governance — Fine-grained authorization controls scope which agents can access which subgraphs, node sets, and tiers. Deny-by-default with audited reason codes.

© 2026 LATAM Airlines

- Document Extraction Pipeline — Ingest PDFs and PowerPoint files, extract entities, deduplicate, and populate the Semantic graph automatically — turning institutional documents into queryable organizational memory.

3. Architecture Brief

AMG is a GCP-native Python service built on FastAPI, deployed on Cloud Run, backed by Google Cloud Spanner Graph for the memory store. All components are containerized and managed through Terraform infrastructure-as-code. The system deploys as a dedicated single-organization service within the customer's GCP project — no data ever leaves the customer's environment.

3.1 Memory Tiers

AMG divides all knowledge into three strictly isolated tiers with different retention policies, mutability rules, and access controls. The design mirrors human memory architecture: working memory (Context), long-term memory (Semantic), and autobiographical log (Audit).

Tier	Scope & Owner	What It Stores	Updated By	Lifecycle
CONTEXT	Per user / per session	User-specific conversation turns, agent responses, session state, intermediate reasoning. Scoped strictly to one session and one user — no cross-user sharing.	Every message sent or received, session start/end events, agent tool calls during a turn.	24–48 hours (rapid Bayesian decay on idle sessions)
SEMANTIC	Organization-wide	Durable facts, entity relationships, discovered insights, ingested document knowledge. Shared across all agents in the organization within ABAC policy scope.	Agent fact additions, document extraction pipeline, DataHub ingestion, manual knowledge inserts via API.	Months to years (slow decay; Bayesian reinforcement on repeated access)
AUDIT	Permanent / compliance	Every memory access, tool call, authorization grant/deny, reasoning step, and orchestration trace. Indexed with reason codes.	Append-only — written by every operation across all tiers. Never modified or deleted.	Permanent retention

> Privacy note: The Context tier is the right home for any user-identifiable data in agent interactions (passenger IDs, employee IDs, session content). Its 24-48h decay and strict per-session scope mean personal data is not persisted indefinitely — aligned with data minimization principles.

3.2 MindMap Reasoning Pipeline

© 2026 LATAM Airlines

The MindMap pipeline is the core reasoning engine of AMG. It is inspired by the research paper "MindMap: Knowledge Graph Prompting Sparks Graph of Thoughts in Large Language Models" (Wen, Wang & Sun, ACL 2024 — github.com/wyl-willing/MindMap), which demonstrated that LLMs grounded in knowledge graph evidence produce structurally coherent, evidence-backed responses far more reliably than free-form generation or naive RAG. AMG adapts these ideas into a production-grade, enterprise-safe, deterministic pipeline.

The pipeline enforces a strict evidence discipline: no LLM generates answers directly from user input. Every response passes through four deterministic stages:

Stage	What It Does	Output
1. Mine	Extracts structured intent and named entities from the agent request. Identifies query anchors for graph traversal.	Entity list + intent classification
2. Aggregate	Traverses the Spanner Graph from identified anchors. Bounded traversal depth and cognitive weight thresholds filter out stale or low-confidence paths. Only ABAC-permitted subgraphs are accessed.	Evidence path set (typed graph traversal results)
3. Serialize	Converts graph paths and edges into canonical, traceable fact payloads. Each fact receives a unique ID for citation in the final response. Provenance (source node, edge type, confidence score) is preserved.	Serialized fact payload with fact IDs
4. Synthesize	Generates the agent response ONLY from the serialized fact payload using the configured LLM (Cosmos GenAI Gateway). Freshness and confidence gates are checked before synthesis. If no facts exist or confidence is below threshold, returns an explicit insufficiency signal — never guesses.	Grounded response or explicit insufficiency

> Grounding validation: After synthesis, AMG verifies that every claim in the response references a fact ID present in the serialized evidence. Any claim referencing an unknown fact ID causes the response to fail closed — a hard guarantee against hallucination.

3.3 Cognitive Weighting System

The Cognitive Weighting system gives AMG its ability to deliver high-accuracy retrieval at low token cost. It is inspired by Engram — an open-source provable memory infrastructure for AI agents (github.com/Harshitk-cp/engram) that scores 91.4% on LongMemEval, the ICLR 2025 benchmark for long-term conversational memory. AMG adapts Engram's core insight — that memories should gain confidence when reinforced, lose it when contradicted, and fade when unused — into a Bayesian log-odds weight model native to the Spanner Graph data model.

Every node and edge in the memory graph carries a LogOddsWeight:

© 2026 LATAM Airlines

Mechanism	Formula / Rule	Effect
Initial weight	LogOddsWeight = 0.5 (62% confidence baseline)	New facts enter with moderate confidence — not assumed fully reliable.
Bayesian reinforcement	Weight += ln(1.25) ≈ +0.22 bits per successful access	Frequently retrieved facts gain confidence automatically. High-signal knowledge rises to the top of traversal results.
Temporal decay	Weight(t) = Weight(0) × e^(–0.05 × hours_since_access)	Unused facts fade out. A fact at weight 2.73 after heavy use decays to near-zero after one week without access — mimicking forgetting.
Pruning threshold	Weight < 0.30 → soft-deleted from active traversal	Stale, low-confidence facts stop polluting retrieval results without a hard delete.
Contradiction detection	On ingest, conflicting facts are flagged with a tension marker	Synthesis is gated: contradicted facts surface as ambiguous, not authoritative.
Freshness gating	Synthesis checks average_freshness of evidence set against freshness_threshold	Responses explicitly signal when evidence is stale — preventing overconfident answers from outdated knowledge.

The direct business impact of this system: instead of sending hundreds of loosely relevant RAG chunks to the LLM, AMG sends only the high-confidence, fresh, weighted facts that passed all filters. This bounds the synthesis context window, reduces LLM token consumption per agent turn, and raises answer accuracy — the two most important cost and quality levers for enterprise AI at scale.

3.4 Core API Endpoints

Endpoint	Method	Purpose
/health	GET	Liveness and readiness probe for Cloud Run health checks
/api/v1/agent/respond	POST	Full multi-tool agent turn: orchestrate memory recall, graph search, evidence synthesis, and trace logging in a single request
/api/v1/mindmap/execute	POST	Direct MindMap pipeline execution — Mine → Aggregate → Serialize → Synthesize
/api/v1/mcp/invoke	POST	MCP tool invocation: memory_search, memory_add_fact, memory_recall_context, memory_log_trace. Compatible with Claude Code, Codeen, and the Cosmos GenAI Hub MCP registry.
/internal/decay-weights	POST	Scheduled cognitive weight decay — reduces weights for idle nodes/edges

© 2026 LATAM Airlines

Endpoint	Method	Purpose
/internal/prune-weights	POST	Remove soft-deleted low-confidence evidence from Semantic tier

3.5 Technology Stack

Component	Technology	Notes
Runtime	Python 3.14 / FastAPI	Strict type-checking (basedpyright, 0 errors, 0 warnings)
Memory Store	Google Cloud Spanner Graph	ISO GQL queries, interleaved edge storage for fast traversal
Deployment	Cloud Run (containerized)	Terraform IaC, 3-environment strategy (dev/intg/prod)
Observability	GCP Logging + Tracing + Metrics	Structured JSON logs, OpenTelemetry traces, Cloud Monitoring
Auth & Governance	Internal ABAC or External Delegation	Pluggable: connect to Okta, Auth0, Entitlements Platform, or internal policy
LLM Backend	Cosmos GenAI Gateway	Calls claude-sonnet-4-6, claude-opus-4-8, gemini-2.5-pro, gpt-5 per config
CI/CD	GitLab CI / Cloud Build	Pre-commit hooks, ruff linting, basedpyright, pytest — required before merge
Tests	pytest — 843 tests, 100% passing	Unit + Integration + Cross-tenant isolation + Chaos scenarios
Doc Ingestion	PDF/PPTX extraction pipeline	Fuzzy entity deduplication, confidence scoring, deployed on Cloud Run

3.6 Deployment Model

AMG is deployed as a dedicated single-organization service within the customer's GCP organization. All agents in the organization share memory within the same company_id context. There is no multi-tenancy — data never leaks across organizations. The deployment follows a three-environment strategy managed through Terraform workspaces:

- Development — Agent integration testing, MCP wiring, schema validation, first squad onboarding.
- Integration (INTG) — Chaos engineering, load testing (7,500+ req/sec target), cross-tenant isolation validation, compliance documentation review.
- Production — Canary deployment (5% traffic) → 24h monitoring → full traffic. Rollback time: < 5 minutes to previous version.

© 2026 LATAM Airlines

Governance Mode	Description	When to Use
Internal ABAC (default)	Authorization policies stored in Spanner. LATAM-managed. No external dependencies.	Recommended for initial deployment and most use cases
External Delegation	Authorization decisions delegated to an external policy service via POST /authorize. Response: allow/deny with reason code.	When LATAM wants to sync AMG permissions with an existing identity platform (e.g., a centralized entitlements service)

4. Strategic Value for LATAM Airlines

LATAM Airlines has already invested significantly in a modern AI and data platform: the Cosmos GenAI Gateway (accessing Claude, GPT-5, Gemini, Llama models), DataHub for enterprise data governance across domains (CUSTOMER, MAINTENANCE, REFERENCE DATA, DATA INSIGHTS), Backstage for service catalog, and the Cosmos Feedback Engine for chatbot quality improvement. Agent Memory Graph is the persistent coordination layer that makes these investments compound: every agent improves the shared memory; every squad benefits from what others have learned.

4.1 LATAM's GenAI Ecosystem Today

From the Cosmos GenAI Gateway, LATAM operates a diverse model fleet served from Vertex AI and OpenAI: Claude Opus 4.8, Claude Sonnet 4.6, GPT-5, Gemini 3.1 Pro, Llama 4 Scout, and more — with per-user cost tracking and budget controls. LATAM also runs the Cosmos Feedback Engine, a four-module chatbot quality improvement system (Dataflow pipeline, expert review UI, Cloud Run manager, daily settler) that already enriches chatbot responses with human-validated knowledge. The platform demonstrates significant AI maturity.

However, these models are stateless: each call starts from scratch, with no shared context between agents, squads, or sessions. AMG directly addresses this gap — it is the persistent, shared memory layer that transforms isolated model calls into coordinated, context-aware, auditable agent workflows.

4.2 Chatbots Facing LATAM Personnel and Customers

One of the highest-impact applications of AMG at LATAM is as the memory backbone for chatbots and AI assistants that interact with two distinct audiences: internal LATAM personnel and external passengers. Both segments have active AI initiatives and unsolved memory problems.

4.2.1 Internal Personnel Chatbots

LATAM's ADO teams, operations crews, maintenance engineers, and contact center agents increasingly interact with AI copilots for knowledge lookup, process guidance, and decision support. Without persistent memory, these agents reset on every session — forcing employees to re-provide context, re-explain ongoing issues, and re-discover solutions that other agents have already found.

© 2026 LATAM Airlines

- Maintenance (MAPIS) — Technicians querying the MAPIS knowledge base (already cataloged in BigQuery: emx-data-prod-e6c5.raw_emx_mapis_rag.mapis_knowledge_jan_26) benefit from an AI copilot that remembers prior diagnoses, knows which solutions were applied to similar faults, and cites its sources for safety-critical traceability.
- Operations copilots — Disruption management agents share a common Semantic graph so that one agent's finding about a delay cause immediately informs all other coordinating agents — no redundant queries, no contradictory decisions.
- Data governance assistants — ADO engineers interacting with DataHub for dataset ownership, lineage review, and access request triage get a memory-backed assistant that retains project context across weeks of work without re-prompting.
- ADO engineering copilots — Engineers onboarding to new datasets or pipelines get a copilot that remembers what they've explored, what decisions were made, and what patterns exist — cutting onboarding time significantly.

4.2.2 Passenger-Facing Chatbots

LATAM's customer experience domain is one of the most data-rich in the organization: NPS surveys (digital, contact center, and in-flight), customer voice tables, and interaction logs are already cataloged in DataHub under the CUSTOMER domain (cus-data-prod.customer_experience.*). These datasets represent thousands of daily passenger interactions — a rich source for grounding passenger-facing AI.

AMG enables a qualitatively different class of passenger chatbot:

- Session continuity — A passenger who starts a booking inquiry, disconnects, and returns hours later finds the agent still knows their itinerary, preferences, and prior steps — not a blank slate requiring re-explanation.
- Cross-channel consistency — A passenger who called the contact center yesterday and now chats on the app encounters the same context. The shared Semantic tier (scoped per passenger entity) bridges channels without requiring a unified CRM lookup on every turn.
- Grounded, trustworthy answers — Fare rules, baggage policies, and flight statuses are retrieved from fact-grounded graph evidence — not hallucinated from training data that may be months out of date.
- Compliance-ready audit trail — Every passenger interaction is logged in the Audit tier. For regulated complaints, data subject access requests, or quality investigations, LATAM has a complete, immutable reasoning trace for every chatbot decision.
- NPS signal integration — DataHub's NPS and customer voice tables can feed the Semantic tier: agents that know a passenger had a poor recent experience can adapt their tone and escalation threshold accordingly — a personalization layer grounded in real behavioral data.

> Synergy with Cosmos Feedback Engine: AMG's Semantic tier is the ideal long-term store for the validated Q&A; pairs that the Feedback Engine produces daily. Instead of only updating a Cloud SQL vector database, validated answers can be ingested into the AMG Semantic graph — making them available to all agents organization-wide, not just the chatbot the Feedback Engine was built for.

4.3 High-Impact Use Cases Across LATAM Domains

© 2026 LATAM Airlines

Domain	Use Case	AMG Capability Used	Business Impact
Customer Experience	Multi-session passenger chatbot with persistent context across app, web, and contact center channels	Context tier (per-passenger), cross-channel Semantic facts, NPS data ingestion	Reduced handle time, higher first-contact resolution, consistent cross-channel UX
Revenue Management	Agents querying fare rules, market data, and pricing policy without re-ingesting knowledge on every call	Semantic tier, MindMap grounding, cognitive weight prioritization	Faster analysis cycles, zero hallucinated fare rules in automated decisions
Maintenance (MRO)	AI copilot for technicians accessing MAPIS knowledge with safety-critical traceability for every recommendation	Document extraction pipeline (MAPIS PDFs), Audit tier, grounding validation	Compliance-ready AI recommendations with full reasoning audit trail
Data Governance	DataHub-integrated assistant with persistent context about dataset ownership, lineage, access history, and governance status	Semantic tier (DataHub entities), ABAC governance, Audit tier	Automated governance triage with complete auditability
Operations	Multi-agent coordination for disruption management — no agent acts on stale or contradictory information	Shared Semantic graph, contradiction detection, cross-agent context	Consistent, non-redundant operational decisions at scale
Internal Copilots (ADO)	AI assistants for engineers and data scientists that remember project context, architectural decisions, and prior analyses across weeks	Context + Semantic tier, MCP integration with Claude Code / Codeen	Faster onboarding, eliminated context re-explanation overhead

4.4 Alignment with LATAM's Existing Infrastructure

- GCP-Native — Deploys in LATAM's GCP organization with Cloud Run + Spanner + IAM patterns already established across the Cosmos platform.
- Cosmos GenAI Gateway Integration — AMG's Synthesize stage calls the GenAI Gateway as its LLM backend. LATAM's existing Claude/GPT/Gemini access, cost tracking, and rate limits apply.
- MCP Protocol Support — AMG exposes an MCP-compliant tool interface compatible with Claude Code, Codeen, and the Cosmos GenAI Hub MCP registry. Register it as an internal MCP Server and all LATAM teams can discover and use it from the Hub catalog.
- DataHub Integration — AMG's extraction pipeline can ingest DataHub-cataloged assets, enriching the Semantic tier with dataset ownership, lineage, and quality metadata.
- Backstage Catalog — AMG ships with a catalog-info.yaml making it immediately registerable as a Backstage component in the Cosmos service catalog.
- Cosmos Feedback Engine — Validated Q&A; pairs from the Feedback Engine can flow into the AMG Semantic tier, making expert-validated knowledge available to all agents.

4.5 The Organizational Multiplier Effect

© 2026 LATAM Airlines

The compounding value of AMG grows as more agents and squads adopt it. Today each AI initiative at LATAM operates in isolation — a maintenance chatbot knows nothing about what a revenue agent discovered last week. AMG creates a shared organizational knowledge graph where every agent's findings enrich the corpus for all future agents. Cognitive weighting ensures that the most frequently useful facts rise to the top automatically, while unused or contradicted knowledge decays — so the shared memory self-maintains without manual curation.

> Compliance alignment: LATAM operates in regulated environments — aviation safety, financial reporting, passenger data protection. AMG's immutable AUDIT tier and ABAC controls provide the traceability and access governance required for AI use in safety-critical and compliance-sensitive decisions. No simple RAG or stateless chat stack provides this.

5. Development & Deployment Plan for LATAM

AMG v0.2 is production-ready. The LATAM deployment plan focuses on integration, adoption, and scaling — not building from scratch. The following plan is structured across four phases over approximately 16 weeks.

5.1 Required Team

Role	Qty	Responsibility	Suggested Source
Platform Engineer (GCP)	1 FTE	Terraform, Cloud Run, Spanner, IAM, CI/CD setup, monitoring	Cosmos Platform Engineering
Backend Engineer (Python)	1–2 FTE	API integration, GenAI Gateway wiring, Feedback Engine connector, extraction pipeline adapters	GenAI Squad (GENAI) or ADO chapter
Data / AI Engineer	1 FTE	Entity schema design, Semantic graph population strategy, DataHub and MAPIS connectors	Chapter Data & AI (any squad)
Security / Governance	0.5 FTE	ABAC policy design, IAM audit, compliance documentation, data residency confirmation	Information Security / ADO Governance Enabler
Product Manager / Lead	0.5 FTE	Pilot squad coordination, KPI tracking, stakeholder alignment, customer chatbot requirements	Chapter Lead (GENAI squad or ADO chapter head)
QA / Integration	0.5 FTE	End-to-end testing, chaos engineering, cross-tenant isolation validation	Within any of the above squads

Total: approximately 4.5–5.5 FTE across 16 weeks. A lean pilot (1 use case, 1 squad) can start with 3 FTE in Phase 1.

© 2026 LATAM Airlines

5.2 Estimated Budget

Category	Item	Est. Monthly (USD)	Notes
Infrastructure	Cloud Run (2 instances, autoscale)	$150–400/mo	Scales with request volume
Infrastructure	Cloud Spanner (1 node, ~10k RU/day)	$365/mo	Scales horizontally; 1 node handles hundreds of req/sec
Infrastructure	Cloud Build / CI pipeline	$50–100/mo	Based on build frequency
Infrastructure	GCP Logging + Monitoring + Tracing	$50–200/mo	Depends on log volume and retention
LLM Inference	Cosmos GenAI Gateway (Synthesis stage)	Existing budget	Claude Sonnet 4.6: ~$0.003/1k input tokens via gateway
Engineering	Platform team (4.5 FTE × 16 weeks)	One-time ~$120–180k	Leverages internal engineers; no vendor licensing
	TOTAL INFRA (monthly, steady-state post go-live)	~$615–1,100/mo	

> Cost note: At pilot scale (< 1k req/day), a fractional Spanner instance reduces infrastructure to ~$100–200/mo. Cognitive Weighting bounds LLM context per request, keeping inference cost predictable at scale.

5.3 Phases & Timeline

Phase 1 — Foundation & Pilot (Weeks 1–4)

Goal: Deploy AMG in LATAM's GCP dev environment and validate the core memory pipeline with one real use case (recommended: MAPIS maintenance chatbot or DataHub governance assistant).

Week	Activity	Owner	Deliverable
1	GCP org setup: Spanner, Cloud Run, IAM, Terraform workspaces, monitoring alerts	Platform Engineer	Dev environment live
1–2	Cosmos GenAI Gateway wiring to AMG Synthesize stage; smoke-test end-to-end grounded response	Backend Engineer	Full pipeline returning grounded response
2	Backstage catalog-info.yaml registration; MCP registration in GenAI Hub	Platform Engineer	AMG visible in Cosmos Backstage + discoverable from Claude Code / Codeen
2–3	First pilot squad onboarding — GenAI squad with one use case; 50+ agent interactions in dev	Product Lead + Backend	Pilot validated in dev

© 2026 LATAM Airlines

Week	Activity	Owner	Deliverable
4	Phase 1 review: grounding quality, P99 latency, token cost vs. baseline	All	Go/No-Go sign-off for Phase 2

Phase 2 — Integration & Data Enrichment (Weeks 5–9)

Goal: Enrich the Semantic graph with LATAM organizational data, integrate DataHub and the Feedback Engine, and expand to 2–3 pilot squads including one passenger-facing chatbot.

Week	Activity	Owner	Deliverable
5–6	DataHub connector: ingest dataset metadata, ownership, lineage into Semantic tier	Data/AI Engineer	DataHub entities queryable from AMG
6–7	Document extraction pipeline for MAPIS and LATAM internal knowledge bases	Backend + Data Engineer	First knowledge corpus ingested and weighted
7	Cosmos Feedback Engine connector: validated Q&A; flows into Semantic tier	Backend Engineer	Feedback Engine knowledge available to all agents
7–8	ABAC policy design aligned with LATAM IAM / identity model	Security / Governance	Authorization policy document signed off
8–9	Expand to 2–3 squads: customer experience chatbot pilot + one internal squad	Product Lead	Multi-squad pilot live with passenger-facing use case
9	Phase 2 review: data coverage, auth compliance, coordination between squads	All	Sign-off for Phase 3

Phase 3 — Hardening & Production (Weeks 10–14)

Goal: Deploy to INTG and PROD, complete compliance documentation, and certify for safety-critical and regulated use cases.

Week	Activity	Owner	Deliverable
10–11	INTG deployment: chaos engineering, load testing (7,500+ req/sec), cross-tenant isolation validation, NPS data integration test	Platform + QA	INTG environment validated

© 2026 LATAM Airlines

Week	Activity	Owner	Deliverable
11–12	Compliance documentation: audit trail review, ABAC policy audit, passenger data residency confirmation, GDPR/PDPA alignment check	Security / Governance	Compliance sign-off document
12–13	PROD canary deployment (5% traffic) → 24h monitoring → full traffic	Platform Engineer	PROD environment live
13–14	Runbook, escalation procedures, on-call setup, SLO dashboard in Cloud Monitoring	Platform + Product	Operational runbook published
14	Phase 3 review: availability, audit completeness, passenger chatbot quality	All	Production sign-off

Phase 4 — Scale & Expansion (Weeks 15–16+)

- Onboard all interested Chapter Data & AI squads with self-service documentation and pre-seeded domain knowledge graphs (maintenance, revenue, customer experience).
- Expand passenger chatbot to additional channels (app, web, contact center) using the shared Context + Semantic tier for cross-channel continuity.
- Evaluate publishing AMG as a reusable Cosmos platform component available to all LATAM product squads via the Backstage catalog.
- Define v0.3 roadmap: embedding-based retrieval, vector search over the Semantic tier, agentic workflow templates, and NPS-driven personalization signals.

5.4 Risk Register

Risk	Likelihood	Impact	Mitigation
Cloud Spanner quota delays	Low	Medium	Request quota increase early in Phase 1; Spanner emulator available for local dev immediately
Passenger data privacy requirements (GDPR/PDPA)	Medium	High	Context tier has 24-48h decay; no persistent PII in Semantic tier by policy; ABAC controls access; legal review in Phase 3
ABAC policy complexity growth	Medium	Low	Start deny-by-default for all agents, then grant access incrementally — not the reverse. Keep policies declarative.

© 2026 LATAM Airlines

Risk	Likelihood	Impact	Mitigation
Semantic graph data quality (garbage in, garbage out)	Medium	High	Use DataHub quality scores to gate ingestion; extraction pipeline includes fuzzy dedup and confidence scoring
LLM cost overrun at passenger chatbot scale	Low	Medium	Cognitive weighting bounds context per request; Cosmos Gateway provides per-user budget tracking and hard limits
Team availability (lean team)	High	Medium	Phase 1 can run with 3 FTE; AMG is already built — no core R&D; needed. Scope pilot tightly before scaling team

6. Success Metrics

Metric	Target	Measurement
Grounding Rate	100% of responses backed by evidence or explicit insufficiency	Audit tier: grounded flag per response
Hallucination Rate	0% (no unsupported claims pass the grounding validation gate)	Automated grounding check in synthesis stage + spot audit review
Token Cost Reduction	>30% reduction vs. baseline RAG per agent turn	Cosmos Gateway per-turn token logs before vs. after AMG adoption
P99 Latency	< 50ms end-to-end (current measured baseline: < 5ms)	Cloud Monitoring percentile metrics
Availability	99.9% uptime	Cloud Run health checks + uptime alert in Cloud Monitoring
Audit Coverage	100% of authorization decisions logged with reason code	Audit tier query: granted/denied events per day
Agent Onboarding	5+ LATAM squads using AMG by end of Phase 4	MCP invocation count per squad via GenAI Gateway logs
Passenger Chatbot Quality	NPS improvement ≥ 5 points vs. stateless chatbot baseline	DataHub NPS tables (cus-data-prod.customer_experience_analysis.nps_*)
Cost per Request	< $0.01 per agent turn (infra + LLM)	Gateway cost logs + Spanner usage metrics combined

7. Recommendation & Next Steps

Agent Memory Graph is production-ready today. The recommended approach for LATAM is to begin with a focused 4-week pilot led by the GenAI squad on a single high-value, well-scoped use case — either the

© 2026 LATAM Airlines

MAPIS maintenance knowledge assistant or the DataHub governance copilot. Both have clear data sources already cataloged, clear quality metrics, and limited passenger data privacy implications, making them ideal for a first validation before expanding to external-facing chatbots.

- Week 1: Platform engineer deploys GCP dev environment; GenAI Gateway wired to AMG.
- Week 2: GenAI squad runs first agent interactions through the MindMap pipeline; validate grounding and audit trail completeness.
- Week 3: Security/governance stakeholder reviews ABAC configuration and audit logs.
- Week 4: Go/No-Go decision. If pass thresholds → proceed to full 16-week plan. If not → identify root cause and iterate before expanding.

The passenger-facing chatbot use case — the highest-scale and highest-impact application — should enter the plan in Phase 2, after the internal pilot has validated the core pipeline and governance model. This sequencing manages risk while preserving the timeline for customer-facing value delivery.

> Suggested contact point: GenAI squad (Squad Lead: claudio.urbina@latam.com), in coordination with the ADO Chapter Data & AI Ops (Chapter Head: grp_llmops_team@latam.com). For governance alignment: ADO Governance Enablers (juanjose.gutierrezt@latam.com). For passenger chatbot requirements: Chapter Data & AI Passenger (ATO-ATW).

© 2026 LATAM Airlines